Which companies use employment subsidies for young workers?

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Abstract

Active labour market policies like employment subsidies require the involvement of employers to be implemented. However, little is known about the types of firms that participate in these programs. This article merges administrative data of subsidy recipients of a program targeted to young workers from Andalusia (Spain) with balance sheet data of Spanish non-financial companies obtaining a sample of 115,867 units. In a first stage, we avoid modelling assumptions by estimating summary statistics on the socioeconomic features of the companies that used the subsidy. In a second stage, to measure the associations with flexible models, we estimate generalized additive models with restricted cubic splines. The model estimates indicate that there are two variables with outstanding prediction ability of participation in employment subsidies: the temporary rate of the workforce and the gross value added of the firm. Thanks to our flexible model, we detect that the relationship between using a subsidy and the temporary rate is non-linear, specifically hump-shaped. Considering interquartile variations in the covariates, a rise in the temporary rate or the gross value added is associated with a substantial increase in the odds of getting an employment subsidy for young people. Our results show the usefulness of generalized additive models to shed light on the actual recipients of the subsidies. This may be of interest when policymakers expect employers to informally train the workers or if they want to foster innovation or certain economies of scale.

Keywords: employment subsidies, active labour market policies, public policy.

JEL codes: J68, J08.

MSC2020 codes: 62P20, 62P25.

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Título: ¿Qué empresas usan las subvenciones al empleo para jóvenes?

Resumen. Políticas activas de empleo como las subvenciones al empleo requieren la participación de los empresarios para su implementación. Sin embargo, sabemos poco sobre los tipos de empresas que participan en estos programas. Este artículo fusiona datos administrativos de los beneficiarios de subvenciones de un programa dirigido a jóvenes de Andalucía (España) con datos de balances de empresas españolas no financieras obteniendo una muestra de 115.867 unidades. Primero, evitamos las asunciones de modelización estimando estadísticos de resumen sobre las características socioeconómicas de las empresas que utilizaron el subsidio. Segundo, para medir las asociaciones con modelos flexibles, estimamos modelos aditivos generalizados con *splines* cúbicos restringidos. Las estimaciones del modelo indican que hay dos variables con una capacidad sobresaliente de predicción de la participación en subvenciones al empleo: la tasa de temporalidad de la plantilla y el valor añadido bruto de la empresa. Gracias a la flexibilidad de nuestro modelo, detectamos que la relación entre la utilización de una subvención y la tasa de temporalidad no es lineal, teniendo forma de joroba. Considerando las variaciones intercuartílicas de las covariables, un aumento de la tasa de temporalidad o del valor añadido bruto se asocia a un aumento sustancial de la probabilidad de obtener una subvención al empleo para jóvenes. Nuestros resultados muestran la utilidad de los modelos aditivos generalizados para ilustrar los beneficiarios reales de las subvenciones. Esto puede ser interesante cuando los legisladores esperan que los empresarios formen informalmente a los trabajadores o cuando quieren fomentar la innovación o economías de escala.

Palabas clave: subvenciones al empleo, políticas activas de empleo, política pública.

1. Introduction

Governments are constantly giving public money to private firms to foster the economic activity, but this money is usually difficult to track. The large number of subsidies granted to different companies with different objectives complicates the analysis of these transfers. Although governments usually offer public data on the grant recipients, the interoperability of datasets to study the features of such firms is often limited. However, to facilitate policy evaluation, labour market researchers have highlighted the necessity of understanding the characteristics and motivations of employers to participate in active labour market policies (Bredgaard, 2018; Ingold & Stuart, 2015). Detecting which type of companies receive employment subsidies is a first step in this line.

In this article, we want to know which features of the companies are associated with a larger probability of participating in employment subsidies for young workers. Our objective is to describe how different are the employers that participate and do not participate in these programs. We are especially interested in the economic situation of the companies, since this might give important clues on the persistence of the employment situation after the subsidy. To do that, we study a program implemented in 2017 in the region of Andalusia (Spain) called Bono de Empleo Joven. This employment subsidy is made up of 400 € per month during one year for private hirings of young people with at least lower secondary education.

The empirical approach to answering our research question is based on the detection of participants by joining administrative datasets. Specifically, we match microdata on non-financial firms collected by Banco de España (2023a) with microdata on subsidies provided by the Government of Andalusia (2021), with the help of establishment data offered by Informa & Bureau van Dijk (2021). To analyse the resulting dataset, we have performed both bivariable and multivariable analyses. Firstly, the bivariable stage summarises the differences between participant and nonparticipant firms without imposing any modelling assumption. This allows us to detect which variable on its own presents a larger gap among participants and nonparticipants. Secondly, in the multivariable stage we estimate a generalized additive model with weak assumptions. Using restricted cubic splines, we allow the association of continuous covariates to have a non-linear functional form. This analysis indicates which factor incorporates more non-redundant information about the participants.

The results depend on the quantity of interest, since it seems that there is an association between the firm characteristics studied. When considering only one variable at a time, there is a larger prevalence of participants inside the stratum of medium-sized companies. Interestingly, the relationship between getting the subsidy and firm size is not linear, since microenterprises present almost the same rate of participants than large enterprises. Moreover, participant companies present substantially higher debt ratios and slightly larger profitability ratios than non-participant companies. When considering more than one variable at a time, i.e. when we wish to discount the association between covariates, the story is more nuanced. Considering inter-quartile variations in the continuous covariates, the variables that best predict participation in employment subsidies are the temporary rate of the workforce and the gross value added. In the subsequent analysis, we detect that the association between getting the subsidy and temporary rate is concave. This suggests that there is an important group of participants that generates a large value added, but using strategies of hiring such as a remarkable external flexibility and the use of contributed income to sustain labour costs.

The contributions of our article are related both to the research question and to the data used. Firstly, research had been conducted on the features of the employers or workers that participate in vocational training (Bentolila et al., 2020; Im & Shin, 2022), but not sufficiently in active labour market policies like employment subsidies. This paper addresses a gap in the literature by focusing on an economy in which half of the budget of active labour market policies is spent on employment subsidies (European Commission, 2020). Secondly, evaluations of these policies

typically use panel data on employees and do not add any information on the firm of the employment contract. Although we do not estimate the effect of the program, we look at the employer side of the relationship for the first time in Spain by leveraging information on the companies that received employment subsidies.

The work continues in Section 2 with a review on the literature about the participation of firms in active labour market policies (hereinafter, ALMP) and on the design of our program of interest. In Section 3, the procedures of data collection and analysis are described. Section 4 presents the findings of our study, separating those results that did and did not use a statistical model. Lastly, we conclude with a discussion of the results and future venues of research.

2. Employment subsidies for young workers

In this section, we review the evidence available on the participants in employment subsidies or in similar programs and present the program design of our case of study. The idea is to situate the program in a broader context both theoretically and empirically.

2.1. Theory and use

Inside ALMP, employment subsidies affect the potential productivity of the subsidized worker by decreasing the labour cost for the employer. In general, some theorists have highlighted that these policies have difficulties to create new jobs (vacancies), what we call the distributive hypothesis. From the distributive approach, the tool does not foster new vacancies, but simply changes their recipients (Fay, 1996)[1]. It may be seen as an instrument against inflation (Rehn, 1990), as a subsidization of informal on-the-job training (Wolff & Stephan, 2013) or as a screening device against incomplete information (Crépon & Van Den Berg, 2016). Other theorists have suggested that employment subsides may increase the number of jobs, what we call the incrementalist hypothesis. From the incrementalist approach, ALMP make "vacancies become filled more quickly, they become less costly to firms, and more vacancies are opened" (Calmfors, 1994, p. 13). These positions are not necessarily contradictory, since one situation might predominate at the short run and the other at the long run. In any case, these economic theories reflect the intended goal of the tools, but their actual users might not align with these intentions. To summarize the previous evidence on the companies participating in employment subsidies, we are going to talk first on structural features and second on conjunctural characteristics of the firms.

Structural features of the firm like the sector or the size category barely change from one year to another. Regarding the category of firm size, it seems that there is a consensus about the larger probability of employment subsidy reception of bigger companies (Bellmann & Stephan, 2014; Bishop & Montgomery, 1986; Efstathiou et al., 2019; Huttunen et al., 2013). This seems reasonable due to the greater professionalization of the recruitment process in large enterprises. Bishop & Montgomery (1986) also proposes as explanation the fixed cost that employers have to pay to deal with the program bureaucracy. If the company plans to hire many workers, as might be the case in large firms but not in small ones, the unitary cost decreases. In the case of Spain, an employment subsidy program targeted to young workers in the Basque Country also registered a greater proportion of large firms in comparison to the weight they have in the full population (De la Rica et al., 2022). Concerning the sector of the firms, Bellmann & Stephan (2014) found a larger participation of companies in the health/social work activities and in the education industry for Germany. This makes sense with the idea of having greater presence of labour-intensive sectors (Cronert, 2018). Nonetheless, with a more aggregated categorization, Bishop &

[1] Notice that this point of view assumes that there is always an experienced and a non-experienced worker available for the same job.

Montgomery (1986) found that the construction industry was the sector with the lowest participation in USA.

Conjunctural features of the firm like the profitability or the debt ratio might be more affected by the economic cycle, so we place them in another box. These analyses are more limited due to the rare use of firm-level data in ALMP evaluations. On one side, there is some evidence indicating that participant firms have a relatively bad position in the product market at least at the participation time. Efstathiou et al. (2019) found that those companies that declared to suffer a severe competition had higher probability of getting a subsidy in Luxembourg. The same is found for those that asserted that the demand volatility made them decrease their activity. In a similar vein, in Denmark, Bredgaard & Halkjær (2016) detected a positive association between the participation in employment subsidies and the worsening of the economic situation of the firm. Huttunen et al. (2013) reports that those firms that got the subsidy relatively presented lower salaries in Finland. The evidence for Spain is very scarce, but points to a different direction with similar public policies. In the case of vocational initial training with a dual design, Bentolila et al. (2020) detected that the enterprises that participated were more profitable and presented higher value added than those outside the program.

Researchers have also proposed other variables related to the program design or implementation that might explain program participation. The evidence we have on this matter is scarce, so this forces us to consider all types of employment subsidies as a whole. However, there are substantially different program designs in features like the subsidy amount or the target group that might influence participation (Cronert, 2019). Additionally, other studies like Bishop & Montgomery (1986) have highlighted the importance of marketing actions carried out by the funders of the subsidy (public administrations) to increase the knowledge of the policy.

2.2. Program design

In this article, we study a program called Bono de Empleo Joven (hereinafter, BEJ)[2] implemented during 2017 in the region of Andalusia (Spain)[3]. We concentrate on the version of the program that was regulated in the Ley 2/2015, de 29 de diciembre, de medidas urgentes para favorecer la inserción laboral, la estabilidad en el empleo, el retorno del talento y el fomento del trabajo autónomo (2016). The program design is presented as unstructured data, specifically as text. Therefore, to describe the program in a replicable and standardized way, we have followed the coding instructions for employment subsidies proposed in Junquera (2024). These instructions are based on the literature and highlight three sets of features of the design: target groups, implementers, and tools[4].

Let us start with the agents involved in the program. Regarding the target groups, the intervention is aimed at workers aged 18 to 29 who have at least lower secondary education. The employers of those workers must be organizations at the private sector, but they may or may not be for-profit. They are equally available for companies and for self-employed people who wish to hire a young worker. Concerning the implementers of the program, in this intervention the employers are also those who apply and manage the money assigned for the subsidies. The first stage is the application of the employers, then the public employment service checks the eligibility of the applicants and finally the subsidies are granted until the budget is exhausted. Interestingly, this

[2] It might be translated to English as «Young Employment Voucher».

[3] Different versions of the program were available from 2015 to 2019. As we will argue later, data availability drove us to focus on the 2017 implementation.

[4] As supplementary material, we offer the ATLAS.ti file of the project to track the specific part of the text in which the design feature is regulated.

program does not rank the applicants according to certain criteria, but the funds are allocated based on a first-come, first-served principle.

Let us describe now the tools of the intervention, i.e. the features of the specific actions that should provoke a cognitive or behavioural change in the target groups. The first element we must consider is that program participation is only made up of the employment subsidy, since it does not force the participation in additional ALMP. Such subsidy is defined as an absolute amount of 400 € per month during a total of twelve months. Unlike other program designs, the minimum duration of the contract coincides with the duration of the subsidy, so firms are not forced to preserve the job after subsidy completion. The contracts must be either full-time or part-time, with half of the subsidy provided in the latter case. In any case, there are two limitations on the type of contract: (1) it cannot be a temporary agency work contract and (2) it is not compatible with two types of Spanish employment contracts that offer additional employment subsidies funded by the central government.

3. Data and methods

3.1. Data collection

To answer the research questions, we have combined three longitudinal datasets: the dataset on subsidies provided by the Government of Andalusia (*Base de Datos de Subvenciones de la Junta de Andalucía*, SJA)[5], publicly owned microdata on non-financial corporations and groups offered by the Bank of Spain (*Central de Balances Integrada*, CBI), and privately owned microdata on companies provided by Informa and Bureau van Dijk (*Sistema de Análisis de Balances Ibéricos*, SABI). The resulting dataset includes both those companies with fiscal residence in Andalusia and those with fiscal residence in another region but with at least one establishment in Andalusia[6]. The first subgroup of firms was easily identified in the SJA dataset and then flagged in the CBI dataset. The second subgroup of firms was detected through SABI. In all cases, all the extracted information came from the CBI dataset.

The focal variable is the participation in the BEJ treatment in 2017. Two clarifications are required on the treatment. Firstly, the CBI dataset only includes information on private organizations established as juridical persons (such as societies or foundations), but not on physical persons. In any case, we can see in Table 1 that an important proportion of recipient firms are available in our sample. Secondly, we concentrated on the 2017 program because previous implementations were not affected by the legal requirement of publishing the subsidy recipients in the SJA database. For this reason, we cannot be sure that the dataset includes all the participants before 2017. Posterior implementations of the program decreased substantially the number of participants in relation to the population size. This rareness of the event of interest complicates the use of models with a high number of parameters due to the risk of overfitting (Riley et al., 2019). In any case, the Appendix offers the results for the program participation in 2016, which are consistent with the main findings presented in Section 4.

[5] The database extraction was carried out in December 2021.

[6] To be exhaustive, it should also include those companies with fiscal residence outside Spain but with at least one establishment in Andalusia. Unfortunately, our datasets do not allow us to detect those firms.

Table 1 – Subsidies by year and firms that received at least one subsidy by type of firm and year of implementation.

	2015	*2016*	*2017*	*2018*
Subsidies	1082	3392	4545	2393
Firms	594	2095	2743	1241
* Physical persons	219	790	1039	479
* Juridical persons	375 (63.1 %)	1305 (62.3 %)	1704 (62.1 %)	762 (61.4 %)
** In our quality sample	361 (96.3 %)	1004 (76.9 %)	1348 (79.1 %)	725 (95.1 %)

Note: The quality sample regards the subset of observations that passed the quality checks of the Bank of Spain.

We consider the following covariates: firm size, age of the firm, having received contributed income, gross value added, profitability ratio, debt ratio, unitary labour cost, and temporary worker rate. The definitions of the accounting variables are established by the Bank of Spain, which curates the equivalence of accounting concepts across years. Gross value added at factor cost is the result of subtracting the input value (including taxes, but not labour costs) to the output value (including subsidies). The profitability ratio is the ordinary return on investment (ROI), which is defined as the ordinary net profit divided by the total of assets (both equity and liabilities). The debt ratio is the division of the interest-bearing borrowed funds by the interest-bearing liabilities. More information on these concepts is available at Banco De España (2022, 2023). Notice that the selection of variables has followed previous works to foster the comparability of results across similar public policies (Bentolila et al., 2020).

3.2. Data analysis

Once we have constructed our dataset, we start the analysis considering that our objective is describing the features of the companies that used BEJ subsidies. The data analysis is executed in two stages. In the first step, we estimate summary statistics through bivariable analyses to characterize differences between participant and non-participant firms. In the second step, we estimate multivariable regression models. In all cases, we limit ourselves to the estimation step and do not carry out inferences. The reason is twofold. First, the sampling of companies follows a non-probability process, since firms are self-selected by unknown causes. All firms must provide their balance sheets, but some of them do it written by hand and after the deadline. This makes them missing units in our sample. Second, the sample seems sufficiently large to be interesting even without establishing hard assumptions on the relationship with the population. According to Fernández Cerezo et al. (2024), only considering those companies with their fiscal residence in Andalusia for 2016, the CBI dataset includes the 64.3 % of the population of firms.

In the first section of the analysis, we estimate summary statistics without specifying a model on the data-generating process. We want to check if there are differences in certain economic variables of the firm according to its treatment status. For categorical covariates, we stratify the sample by the value of such covariate and then estimate the probability of participation conditional on belonging to that stratum. For continuous covariates, we stratify the sample by treatment status. Inside each stratum, we estimate the median for the covariate of interest and then compare the results for both strata. The reason why we use the median and not the mean is to avoid the influence of right-tail outliers typical of large companies. The advantage of not using a statistical model is that we do not restrict the possible states of nature of the phenomenon of interest through parametric assumptions (Robins & Greenland, 1986). The disadvantage is that

we need to look at certain parts of the marginal and conditional distributions, which does not give a full picture of the story and overlooks the association between covariates.

In the second section of the analysis, we estimate multivariable models. This increases the number of assumptions with the advantage of shedding light on elements not previously considered. Our target parameters are conditional odds ratios, which are defined for each focal covariate $\tilde{X}$ and each pair of levels x_U and x_L[7]. This parameter collects the magnitude of the association between the outcome Y and the focal covariate $\tilde{X}$ once we account for the association between each $X \in \boldsymbol{X} \setminus \{\tilde{X}\}$ and Y. We denote the conditional odds ratio as $\theta\left(x_U, x_L, \boldsymbol{X} \setminus \{\tilde{X}\}\right)$ and define it formally as

$$\theta\left(x_U, x_L, \boldsymbol{X} \setminus \{\tilde{X}\}\right) \coloneqq \frac{\Pr[Y = 1|\tilde{X} = x_U, \boldsymbol{X} \setminus \{\tilde{X}\}]}{\Pr[Y = 0|\tilde{X} = x_U, \boldsymbol{X} \setminus \{\tilde{X}\}]} \left(\frac{\Pr[Y = 1|\tilde{X} = x_L, \boldsymbol{X} \setminus \{\tilde{X}\}]}{\Pr[Y = 0|\tilde{X} = x_L, \boldsymbol{X} \setminus \{\tilde{X}\}]}\right)^{-1}$$

Note that in the case of continuous covariates, the number of possible contrasts (changing x_U and/or x_L) is infinite. For this reason, we report two pieces of the quantities of interest in the case of these variables. Firstly, we present in table format the estimates of the odds ratios for an interquartile change in each continuous covariate. Secondly, we show plots for the odds of each covariate in function of the specific value fixed for such variable ($\tilde{X} = x$). This allows the reader to estimate $\theta\left(x_U, x_L, \boldsymbol{X} \setminus \{\tilde{X}\}\right)$ just by taking the ratio of two odds at different values of $\tilde{X}$. Formally, each odds is defined as

$$odds\left(x, \boldsymbol{X} \setminus \{\tilde{X}\}\right) = \frac{\Pr[Y = 1|\tilde{X} = x, \boldsymbol{X} \setminus \{\tilde{X}\}]}{\Pr[Y = 0|\tilde{X} = x, \boldsymbol{X} \setminus \{\tilde{X}\}]}$$

To estimate the quantities of interest, we depart from the assumption of a data-generating process that may be specified through the following generalized additive model:

$$\Pr[Y = 1|\boldsymbol{X}] = \frac{1}{1 + \exp\left(-\left(\alpha + \sum_{j=1}^{J} b_j(X_j \in \boldsymbol{X})\boldsymbol{\beta}_j + \epsilon\right)\right)}$$

In this logistic model, the covariates $X_j \in \boldsymbol{X}$ enter through a basis function $b_j(\cdot)$. In the case of quantitative variables, $b_j(\cdot)$ is the basis function that allows a restricted cubic spline (Harrell, 2015). The restricted cubic splines, also known as natural splines, smooth the linearity assumption by allowing flexible functional forms through piecewise cubic functions. In the case of qualitative variables, $b_j(\cdot)$ is simply the dummy encoding function. $\boldsymbol{\beta}_j$ is the vector of regression parameters, α is the intercept and ϵ is the error term.

[7] For brevity, we have removed the individual subindex i.

4. Findings

4.1. Bivariable analyses

In this section, we report the differences between those companies that did and did not participate in the employment subsidies without using any model assumption.

Let us start with the size, the sector and the stickiness of the company to contributed income. In Table 2, we first stratify the sample by category and then estimate the proportion of companies that did participate in BEJ inside each stratum. Regarding the firm size, we see that the share of participants was the largest at the medium-sized enterprises. Interestingly, it seems that there was not a linear relationship between the size of the firm and the probability of getting a BEJ subsidy. Concerning the sector of the firm, the industries with the highest probability of participation were education; human health and social work; and information and communication. Regarding contributed income, we estimate that having received a grant, donation or legacy during the previous year was associated with a higher probability of participating in BEJ.

Table 2 – Conditional probabilities of qualitative variables by treatment status ($t = 2017$).

	Yes
Firm size	
Micro	0.905
Small	3.406
Medium-sized	4.694
Large	1.227
Sector	
B. Mining and quarrying	0
C. Manufacturing	1.686
D, E. Energy and water supply, sewerage, waste, and remediation act	0.328
F. Construction	0.513
G. Wholesale, retail, motors repair	1.598
H. Transportation and storage	0.582
I. Accommodation and food services	0.978
J. Information and communication	2.659
K. Financial and insurance activities	0
L. Real estate activities	0.139
M. Professional, scientific and technical activities	1.664
N. Administrative and support service activities	1.138
P. Education	2.788
Q. Human health and social work activities	2.692
R. Arts, entertainment, and recreation	1.214
S. Other service activities	1.201
Received a grant, donation, or legacy (in 2016)	
No	1.057
Yes	3.746

Note: Each cell shows the row proportion in percentual points. Firm size is measured according to the EU definition.

Table 3 shows the median for each quantitative variable conditional to being or not being treated. In relative terms, the greatest differences between participants and non-participants companies were related to the debt ratio and the gross value added. In general, firms that participated in BEJ registered a much higher gross value added but presented substantially larger debt ratios. Additionally, the companies that got a subsidy were more profitable. Regarding the conditions of their workers, participant enterprises presented larger unitary labour costs and more than doubled the proportion of workers that were hired with a temporary contract. On the contrary, there was barely any difference between the age of the participant and non-participants firms.

Table 3 – Conditional medians of quantitative variables by treatment status ($t = 2017$).

	Yes	*No*
Age (years)	10	11
Gross value added (thousands of euros)	142.93	26.59
Ordinary return on investment (%)	7.93	2.34
Debt ratio (%)	30.23	1.28
Unitary labour cost (euros)	20,182.56	14,041.29
Temporary ratio (%)	37.95	16.98

Note: Each cell shows the conditional median at $t - 1$ (2016). Currency units are at nominal values.

4.2. Multivariable analyses

We have seen that there are some socioeconomic variables that present differences between participant and non-participants firms, but we do not know yet which factor on its own describes better the participants in BEJ. The multivariable analysis allows us to detect which are the covariates with the largest associations with the probability of getting an BEJ subsidy. Nonetheless, modelling gives us some conclusions at the price of assuming a certain functional form between the variables. To smooth the linearity assumption, we allow non-linearities through the pre-specification of piecewise functions (splines), but we still maintain the assumption on the lack of interactions.

The results are presented in Table 4 with three models that go sequentially adding groups of covariates. Summarizing the findings in non-linear specifications is complicated, because the magnitude of the changes in Y depend on the point of departure in X. For now, we present the variations associated to a change from the first to the third quartile of the covariate and then we show more general results. The estimates of model (1) indicate that, considering only these three covariates, the firm size seems the best predictor of participation in BEJ. According to our results, controlling for the other two factors, the odds of having taken the subsidy are multiplied by 6 if you pick a medium-sized company instead of a micro firm. Another interesting finding from this model is that, *ceteris paribus*, having received a contribution in the previous year is associated to an odds of participation one time larger than the odds of those that did not get it.

The following models include more variables trying to detect which factors add more non-redundant information. The first finding that we extract from models (2) and (3) is that, once you incorporate the balance sheet covariates, firm size does not seem a good additional predictor of BEJ participation. Taking this picture of the interquartile changes in the covariates, gross value added seems the best predictor in model (2), but the temporary rate exceeds its magnitude of the association in model (3). Specifically, taking a company with a 50 % of temporary workers multiplies by 3 the odds of BEJ participation in comparison with a company without temporary workers. The second finding that we observed with these models is that having received a contribution in the last year resists quite well the introduction of other covariates. This indicates that the stickiness to these not-for-profit contributions seems a good descriptor of the participants

in BEJ. This is in line with the results of Bishop & Montgomery (1986), who discovered a big association between getting the subsidy in the past and current participation in USA. More surprisingly, profitability measured through the ROI has a null relationship with the BEJ take-up as shown by the odds ratio of 1 for the interquartile contrast. The third finding is related to the sector dummies that better predict program participation once we account for the rest of covariates. Three industries stand out: information and communication; professional, scientific and technical activities; and education. Comparisons with the literature are difficult due to the use of different sector classifications, but Bellmann & Stephan (2014) also indicates the education and the health/social industries as great recipients of subsidies in Germany.

Table 4 – Odds ratios for 2017 on the association with BEJ participation

	(1)	(2)	(3)
Firm size (ref: micro)			
Small	3.703	1.112	1.064
Medium-sized	6.394	1.313	1.103
Large	1.96	0.516	0.562
Received a grant, donation, or legacy (ref: no)	2.279	1.739	1.585
Age of the firm [18 vs. 4]	0.682	0.574	0.65
Gross value added [110.67 vs. 0]		3.678	2.217
Ordinary return on investment [15.28 vs. -0.92]		1.057	1.01
Debt ratio [54.56 vs. 0]		1.428	1.293
Unitary labour cost [23,825 vs. 0]			0.616
Temporary rate [56.5 vs. 0]			3.034
Sector (ref: G. Wholesale, retail, motors repair)			
C. Manufacturing	0.782	0.729	0.685
F. Construction	0.373	0.436	0.458
H. Transportation and storage	0.339	0.262	0.254
I. Accommodation and food services	0.566	0.533	0.444
J. Information and communication	1.633	1.719	1.877
L. Real estate activities	0.136	0.198	0.378
M. Professional, scientific and technical activities	1.176	1.501	1.871
N. Administrative and support service activities	0.697	0.756	0.686
P. Education	1.636	1.819	1.568
Q. Human health and social work activities	1.583	1.522	1.507
RS. Arts, entertainment, and recreation; other services	0.8	0.955	0.864
Observations	112,808	112,808	72,290
Likelihood-ratio statistic	1262.59	1833.49	1407.96
Degrees of freedom	26	33	37
Pseudo-R^2 of Nagelkerke	0.092	0.133	0.117

Note: Time-variant covariates are measured at $t-1$. Accounting variables are winsorized. Gross value added is measured in thousands of euros, unitary labour costs in euros, age in years, and the rest of quantitative variables in percentage. For quantitative variables, odds ratios compare the odds for the third quartile of the covariate versus the odds for the first quartile of the covariate. All models include dummies for the province of headquarters. There are some categories of the sector variable that are not present in the table due to the lack of sample of participants at those industries.

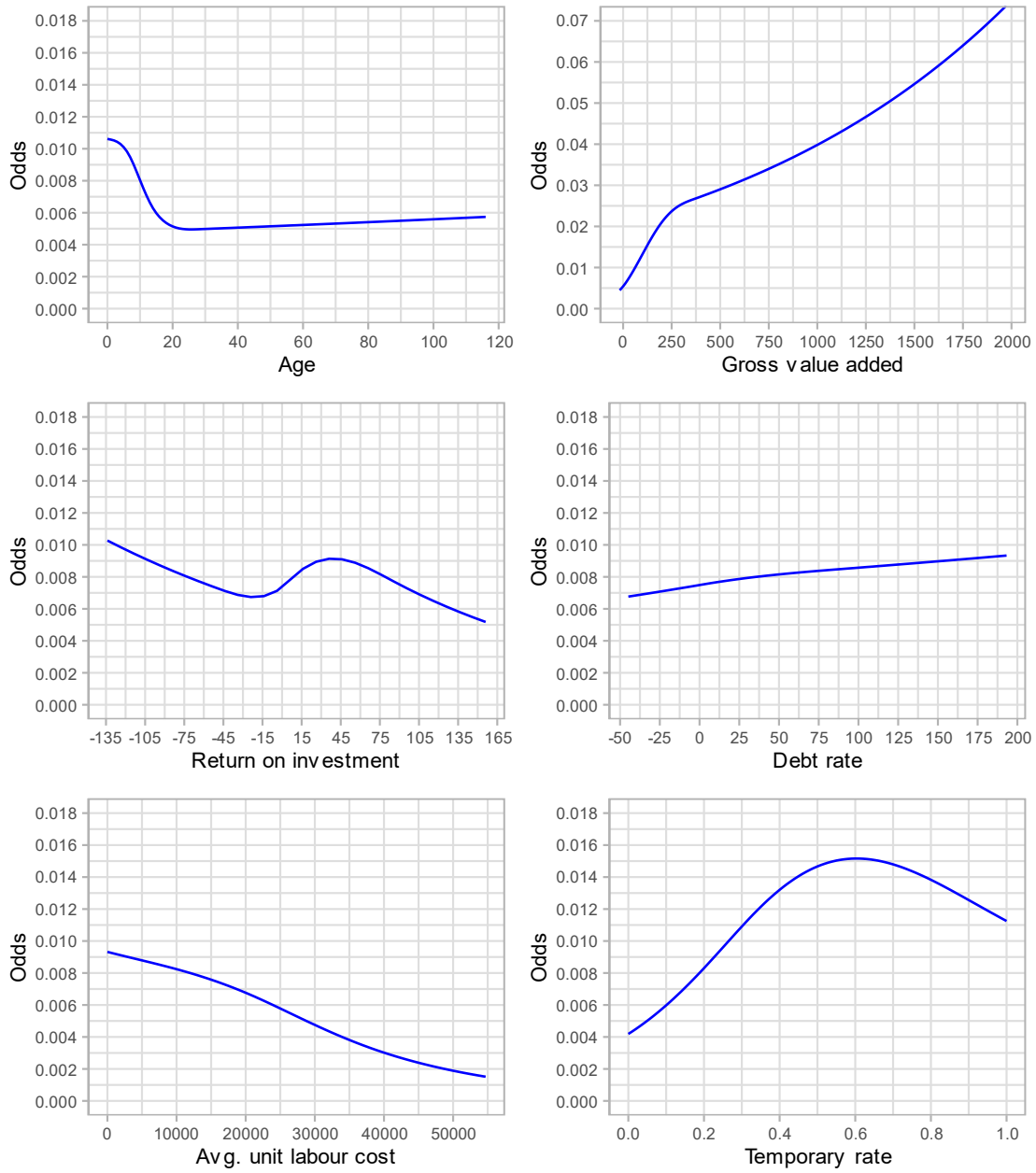
However, one of the contributions of our work is the estimation of non-linearities in the associations, so we should look at the whole dominion of each function. Whereas the interquartile contrasts represent the most representative comparisons, looking at the inner of this range and at the extremes of the distribution might be interesting as well. These non-linearities are easier to understand with plots. Figure 1 shows the odds of getting an BEJ subsidy depending on the value of each covariate included in model (4). To have an easier comparison on the predictive ability of each factor, the scale of the vertical axis is the same for all covariates except for the gross value added[8]. Although we might be initially attracted by the high values that present the odds of this covariate, we must remember that we are mainly interested in the odds ratio. This is the parameter that also considers the association with non-focal covariates, so it gives us an idea of the amount of non-redundant information that each right-hand side variable brings.

Let us start with the covariates that present the largest non-linearities. Regarding the age of the firm, we can appreciate an inverse association for changes in the interval from 0 to 20 years of age and practically a null association for small contrasts beyond the 20 years. For example, the odds of participating in BEJ decrease by approximately half when comparing a company with less than one year of existence to one with 20 years of operation ($\hat{\theta}(20{,}0, \boldsymbol{X} \setminus \{age\}) = 0.05/0.0105$). Concerning the return on investment, i.e. the profitability, we detect that only those variations in the profitability ratio in the interval [-15, +45] euros per euro invested are associated to increases in the odds of participation. In the rest of the domain, increases in the profitability ratio are associated with a decrease in the odds of BEJ participation. A possible explanation about the negative association at the extremes of the market performance might be the lack of knowledge on the program at the left-hand side and the small subsidy amount for those firms at the right-hand side of the distribution. However, these hypotheses would need to be tested in the future. The temporary rate is the third covariate that presents a remarkable non-linear association with the participation in the program. The hump-shaped curve indicates that odds ratios calculated with increases in the temporary rate inside the interval [0, 60] % are associated with rises in the odds of BEJ participation. Interestingly, when the increment in the temporary rate departs from an already high value, we detect a decrease in the odds of getting an EBJ subsidy. The explanation might be related to the program design, since the minimum duration of contracts (12 months) could be too long for a company with a high employee turnover.

The rest of covariates exhibit an almost linear association with the odds of EBJ participation. Consistent with the results for interquartile variations, the gross added value presents a remarkable positive association with the program participation. The relationship is especially strong if we compare firms that barely have a positive gross value added and those that generate around 250,000 € for such quantity, since this variation presents an odds four times bigger ($\hat{\theta}(250{,}0, \boldsymbol{X} \setminus \{GVA\}) = 0.025/0.005$). Also consistent with our expectation, the relationship between the odds of BEJ participation and the average unit labour cost is inverse for the whole dominion. This might be expected due to the design feature of a subsidy with an absolute amount, i.e. a quantity that is independent of the worker's salary. Nonetheless, the relationship between the debt ratio and the odds of getting an EBJ was less expected. The plot reveals an almost negligible association between the two variables, particularly after surpassing the negative debt values. This indicates that there are other variables that describe better the type of companies that participate in employment subsidies for young workers.

[8] We could have drawn all plots using the maximum value of the dependent variable predicted for the maximum value of the gross value added found in our sample. However, there is a remarkable difference between the maximums of the dependent variable attained with the gross value added and the rest of covariates. This would have flattened too much the curves and would have made difficult the visualization of odds ratios generated from different points of departure in the covariate.

Figure 1 – Odds of participation in BEJ in function of the covariates of model (3)



Note: Gross value added is measured in thousands of euros, unitary labour costs in euros, age in years, and the rest of quantitative variables in percentage.

5. Conclusion

In this study, we analyse the economic characteristics of the firms that use employment subsidies to hire young workers using data from administrative and balance-sheet registry sources. Specifically, we focus on a program called Bono de Empleo Joven (BEJ) and implemented in the region of Andalusia (Spain) during 2017. We find, at first sight, that medium-sized companies present the highest proportion of participants in these programs. Nonetheless, once we control for the rest of covariates in a generalized additive model, we discover that firm size is barely associated with program participation. Two covariates outstand as remarkable predictors of using an employment subsidy: the temporary rate of the workforce and the gross value added of the company. Whereas the relationship between the gross value added and getting the subsidy is

practically linear and positive, the association with the temporary rate is clearly non-linear. Thanks to the flexible specification of the model, we detect that variations in the temporary rate of the firm are associated to increases in the odds of BEJ participation until a certain point. If we compare companies with more than 60 % of their workforce with fixed-term contracts, an increase in such variable is related to a decrease in the odds of getting an employment subsidy.

The main limitation of our research has been avoiding the use of statistical inference procedures. We consider that to perform a proper descriptive inference it is necessary to gather more knowledge on the possible selection bias of our estimates. We know that some companies are not in our sample because they delivered late the balance-sheet documents or because they provided them written by hand. Moreover, there might be some important variables correlated with these two reasons of being out of the sample. Therefore, we do not discard selection bias, because these variables might be more present in the population than in our sample. In any case, although we did not test any hypothesis on the population, we employed a sample with more than the 64 % of our sampling frame. Thereby, our results describe at least an enormous number of companies.

Regarding the policy implications of our research, policymakers usually have questions about the effects of the interventions. This is something that has not been answered in our article, but the descriptive evidence may suggest some insights while waiting for credible causal inferences. The first idea we can obtain from our research is the necessity of coordinating labour market and industrial policies when designing employment subsidies. The reason is the diversity of firms involved under these programs in relation to the sector or the firm size. This diversity might not align with industrial policy objectives, since these interventions usually focus on specific sectors or specific sets of firms and use similar policy tools (Criscuolo et al., 2022). The second idea we suggest from our research is the inclusion of clauses in the policy design to avoid the structural use of subsidies by some employers. We have seen that covariates like temporary rate of the workforce or having received contributed income in the previous year do predict well the participation in the current call of the program. This might indicate that some enterprises are using these policies as an *ordinary* source of income. Design features like requiring employment contracts with duration longer than the employment subsidy might repel companies that use these subsidies without intending to keep the subsidised workers.

Future research might explore different research questions or improve the methodological aspects. Concerning the question, we have not studied the effect of treatment participation on outcomes like the employment probability of the worker or the labour productivity of the firm. These questions would deserve further consideration, but they would currently require stronger assumptions due to the nonexperimental treatment allocation. Another slightly different question would be whether employment subsidies with different design features present a different distribution of participant firms. Alterations in design features like the form of the subsidy (e.g., an amount in relative terms), the duration of the contract or a compulsory policy mix with other ALMP might attract different types of companies due to the different costs or benefits attached. Regarding the methods, future work might explore the size of different groups of firms. These groups of firms might be created by imposing certain coincidences in categorical variables or degrees of correlation among continuous covariates through techniques like clustering.

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Appendix

A. Bivariable analyses for 2016

Table A1 – Conditional probabilities of qualitative variables by treatment status ($t = 2016$).

	Yes
Firm size	
Micro	0.618
Small	2.920
Medium-sized	3.841
Large	1.360
Sector	
B. Mining and quarrying	1.398
C. Manufacturing	0.164
D, E. Energy and water supply, sewerage, waste, and remediation act	0.420
F. Construction	1.123
G. Wholesale, retail, motors repair	0.502
H. Transportation and storage	0.725
I. Accommodation and food services	1.741
J. Information and communication	0.000
K. Financial and insurance activities	0.225
L. Real estate activities	1.132
M. Professional, scientific and technical activities	1.013
N. Administrative and support service activities	2.023
P. Education	1.994
Q. Human health and social work activities	0.353
R. Arts, entertainment, and recreation	1.096
S. Other service activities	1.398
Received a grant, donation, or legacy (in 2016)	
No	0.723
Yes	4.360

Note: Each cell shows the row proportion in percentual points.

Table A2 – Conditional medians of quantitative variables by treatment status ($t = 2017$).

	Yes	*No*
Age (years)	9	11
Gross value added (thousands of euros)	181.65	26.65
Ordinary return on investment (%)	9.15	2.36
Debt ratio (%)	27.90	1.33
Unitary labour cost (euros)	20,740.12	14,064.36
Temporary ratio (%)	41.62	17.03

Note: Each cell shows the conditional median at $t - 1$. Currency units are at nominal values.

B. Multivariable analyses for 2016

Table B1 – Odds ratios for 2016 on the association with BEJ participation

	(1)	(2)	(3)
Firm size (ref: micro)			
Small	4.327	1.112	1.064
Medium-sized	6.886	1.313	1.103
Large	3.053	0.516	0.562
Received a grant, donation, or legacy (ref: no)	3.895	1.739	1.585
Age of the firm [18 vs. 4]	0.493	0.574	0.65
Gross value added [120.45 vs. 0.1]		3.678	2.217
Ordinary return on investment [14.52 vs. -0.75]		1.057	1.01
Debt ratio [56.06 vs. 0]		1.428	1.293
Unitary labour cost [24,237 vs. 0]			0.616
Temporary rate [54.1 vs. 0]			3.034
Sector (ref: G. Wholesale, retail, motors repair)			
C. Manufacturing	0.822	0.776	0.774
F. Construction	0.435	0.529	0.542
H. Transportation and storage	0.429	0.346	0.336
I. Accommodation and food services	0.535	0.493	0.402
J. Information and communication	1.583	1.676	1.889
L. Real estate activities	0.291	0.481	0.998
M. Professional, scientific and technical activities	1.16	1.503	1.892
N. Administrative and support service activities	0.869	0.932	0.796
P. Education	1.705	1.846	1.555
Q. Human health and social work activities	1.551	1.453	1.43
RS. Arts, entertainment, and recreation; other services	0.614	0.717	0.635
Observations	105,014	105,014	68,744
Likelihood-ratio statistic	1099.7	1580.6	1320.2
Degrees of freedom	26	33	37
Pseudo-R^2 of Nagelkerke	0.103	0.147	0.136

Note: Time-variant covariates are measured at $t-1$. Accounting variables are winsorized. Gross value added is measured in thousands of euros, unitary labour costs in euros, age in years, and the rest of quantitative variables in percentage. For quantitative variables, odds ratios compare the odds for the third quartile of the covariate versus the odds for the first quartile of the covariate. All models include dummies for the province of headquarters. There are some categories of the sector variable that are not present in the table due to the lack of sample of participants at those industries.

Figure 2 – Odds of take-up in function of different covariates of Model (3) for 2016.

